

LAB Assignment No 9

Distance-Based LED Control using Ultrasonic Sensor with ESP32 and ThingSpeak

In this lab, students will develop an IoT-based distance monitoring system using an ESP32 and an ultrasonic sensor. The system measures the distance of an object and classifies it into three ranges. Based on the detected distance, different LEDs will be activated. The measured distance will also be sent to the cloud using ThingSpeak for real-time visualization and monitoring.

Lab Objectives

- To interface **Ultrasonic Sensor (HC-SR04)** with ESP32
- To measure distance of an object
- To implement multi-level LED control logic
- To send distance data to ThingSpeak cloud
- To visualize distance data using cloud graph

Learning Outcomes

After completing this lab, students will be able to:

- Understand working of ultrasonic sensors
- Measure distance using ESP32
- Implement multi-condition logic (if-else)
- Control multiple LEDs based on sensor data
- Send IoT data to cloud platform
- Visualize and analyze distance data

Required Hardware / Software

Hardware

- ESP32 Development Board
- Ultrasonic Sensor (HC-SR04)
- 3 LEDs (LED1, LED2, LED3)
- $3 \times 220\Omega$ Resistors

- Breadboard & Jumper Wires

Software

- Arduino IDE
- Wokwi Simulator
- ThingSpeak Account

4. Circuit Diagram

Connections:

Component	ESP32 Pin
VCC (Ultrasonic)	5V
GND	GND
TRIG	GPIO 5
ECHO	GPIO 18
LED1	GPIO 16
LED2	GPIO 17
LED3	GPIO 19

◆ 5. Working Principle

1. Ultrasonic sensor sends sound waves
2. Measures echo time to calculate distance
3. ESP32 processes distance and applies logic:

Distance Range	LEDs ON
< 10 cm	LED1
10–15 cm	LED1 + LED2
> 15 cm	LED1 + LED2 + LED3

4. Distance value is displayed on Serial Monitor
5. Data is sent to ThingSpeak every 15 seconds

Wokwi

Program Code

```
#include <WiFi.h>
#include <ThingSpeak.h>

// WiFi credentials
const char* ssid = "Wokwi-GUEST";
const char* password = "";

// ThingSpeak
unsigned long channelID = YOUR_CHANNEL_ID;
const char* writeAPIKey = "YOUR_API_KEY";

WiFiClient client;

// Ultrasonic pins
#define TRIG_PIN 5
#define ECHO_PIN 18

// LED pins
#define LED1 16
#define LED2 17
#define LED3 19

void setup() {
```

```
Serial.begin(115200);

pinMode(TRIG_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);

pinMode(LED1, OUTPUT);
pinMode(LED2, OUTPUT);
pinMode(LED3, OUTPUT);

WiFi.begin(ssid, password);
Serial.print("Connecting");

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("\nConnected!");

ThingSpeak.begin(client);
}

void loop() {
    // Trigger ultrasonic pulse
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH);

    // Calculate distance (cm)
    float distance = duration * 0.034 / 2;

    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm");

    // LED Logic
    if (distance < 10) {
        digitalWrite(LED1, HIGH);
        digitalWrite(LED2, LOW);
        digitalWrite(LED3, LOW);
    }
    else if (distance >= 10 && distance <= 15) {
```

```
digitalWrite(LED1, HIGH);
digitalWrite(LED2, HIGH);
digitalWrite(LED3, LOW);
}
else {
digitalWrite(LED1, HIGH);
digitalWrite(LED2, HIGH);
digitalWrite(LED3, HIGH);
}

// Send data to ThingSpeak
ThingSpeak.setField(1, distance);

int response = ThingSpeak.writeFields(channelID, writeAPIKey);

if (response == 200) {
Serial.println("Data sent to ThingSpeak");
} else {
Serial.print("Error: ");
Serial.println(response);
}

delay(15000); // ThingSpeak delay
}
```

ThingSpeak Setup

- Create a channel
- Add:
 - Field 1 → Distance
- Copy:
 - Channel ID
 - Write API Key

Expected Outputs

□ Simulation (Wokwi)

- Distance changing dynamically
- LEDs turning ON based on distance

- Serial Monitor showing distance

Cloud Output (ThingSpeak)

- Distance graph
- Real-time updates

Lab Tasks

Students must:

- ✓ Interface ultrasonic sensor with ESP32
- ✓ Implement 3-level LED control
- ✓ Display distance on serial monitor
- ✓ Send data to ThingSpeak
- ✓ Observe and analyze cloud graph

Deliverables

- Source code
- Wokwi simulation screenshot
- ThingSpeak graph screenshot
- Brief report

Viva Questions

1. What is ultrasonic sensor?
2. How distance is calculated?
3. What is speed of sound used?
4. Why divide by 2 in formula?
5. What is ThingSpeak?
6. Why delay is required?

```

1  #include <WiFi.h>
2  #include <ThingSpeak.h>
3
4  // WiFi credentials
5  const char* ssid = "Wokwi-GUEST";
6  const char* password = "";
7
8  // ✅ Your ThingSpeak details
9  unsigned long channelID = 3366883;
10 const char* writeAPIKey = "RPN46HCX0M0391KK";
11
12 WiFiClient client;
13
14 // Ultrasonic pins
15 #define TRIG_PIN 5
16 #define ECHO_PIN 18
17
18 // LED pins
19 #define LED1 16
20 #define LED2 17
21 #define LED3 19
22
23 void setup() {
24     Serial.begin(115200);
25

```

```

26     pinMode(TRIG_PIN, OUTPUT);
27     pinMode(ECHO_PIN, INPUT);
28
29     pinMode(LED1, OUTPUT);
30     pinMode(LED2, OUTPUT);
31     pinMode(LED3, OUTPUT);
32
33     WiFi.begin(ssid, password);
34     Serial.print("Connecting");
35
36     while (WiFi.status() != WL_CONNECTED) {
37         delay(500);
38         Serial.print(".");
39     }
40
41     Serial.println("\n✅ Connected to WiFi!");
42
43     ThingSpeak.begin(client);
44 }
45
46 void loop() {
47     // Trigger ultrasonic pulse
48     digitalWrite(TRIG_PIN, LOW);
49     delayMicroseconds(2);

```

```

50    digitalWrite(TRIG_PIN, HIGH);
51    delayMicroseconds(10);
52    digitalWrite(TRIG_PIN, LOW);
53
54    long duration = pulseIn(ECHO_PIN, HIGH);
55
56    // Calculate distance (cm)
57    float distance = duration * 0.034 / 2;
58
59    Serial.print("\ Distance: ");
60    Serial.print(distance);
61    Serial.println(" cm");
62
63    // LED Logic
64    if (distance < 10) {
65        digitalWrite(LED1, HIGH);
66        digitalWrite(LED2, LOW);
67        digitalWrite(LED3, LOW);
68        Serial.println("● LED1 ON");
69    }
70    else if (distance >= 10 && distance <= 15) {
71        digitalWrite(LED1, HIGH);
72        digitalWrite(LED2, HIGH);
73        digitalWrite(LED3, LOW);

```

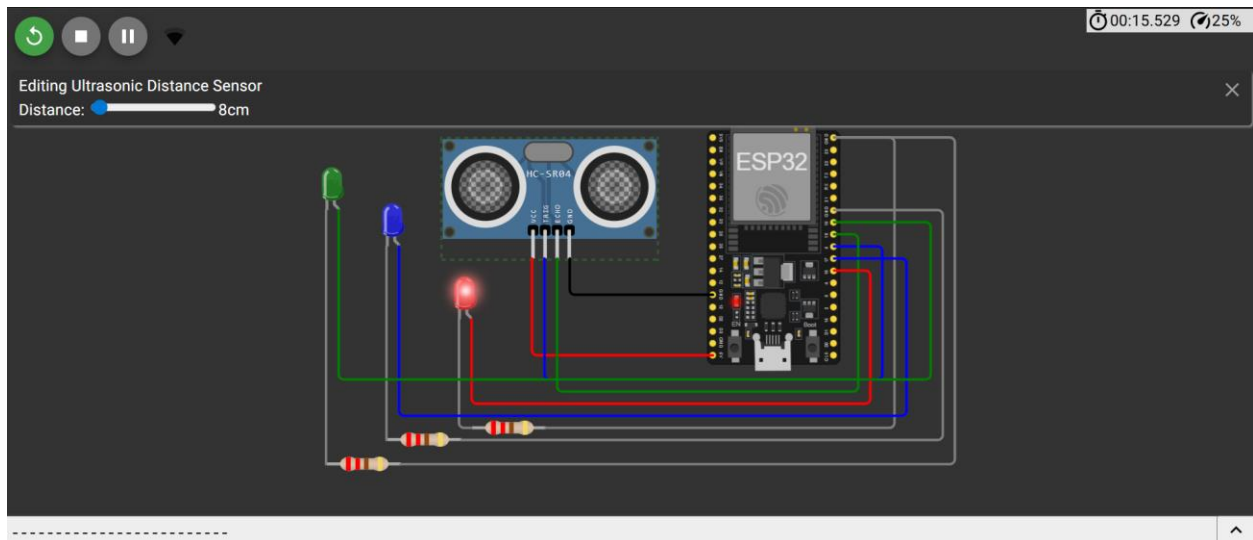
```

74        Serial.println("● LED1 + LED2 ON");
75    }
76    else {
77        digitalWrite(LED1, HIGH);
78        digitalWrite(LED2, HIGH);
79        digitalWrite(LED3, HIGH);
80        Serial.println("● ALL LEDs ON");
81    }
82
83    // Send data to ThingSpeak
84    ThingSpeak.setField(1, distance);
85
86    int response = ThingSpeak.writeFields(channelID, writeAPIKey);
87
88    if (response == 200) {
89        Serial.println("🐦 Data sent to ThingSpeak");
90    } else {
91        Serial.print("✗ Error: ");
92        Serial.println(response);
93    }
94
95    Serial.println("-----");
96

```



```
95     Serial.println("-----");
96
97     delay(15000); // ThingSpeak delay
98 }
```

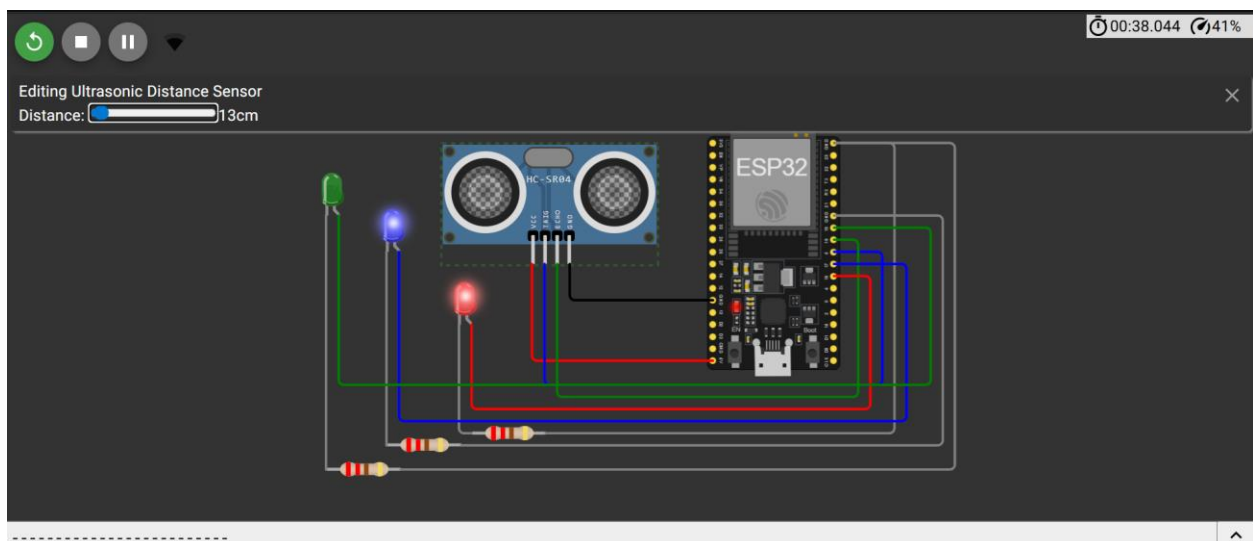


Connecting.....

- ✓ Connected to WiFi!
- Distance: 7.92 cm
- LED1 ON
- Data sent to ThingSpeak

Distance: 7.94 cm

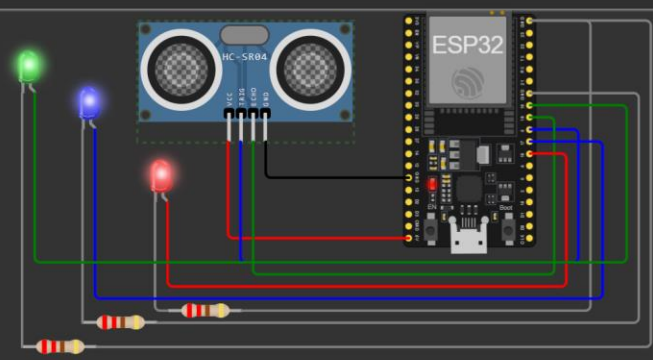
00:38.044 41%



Distance: 7.94 cm
LED1 ON
Data sent to ThingSpeak

Distance: 13.02 cm
LED1 + LED2 ON
Data sent to ThingSpeak

Editing Ultrasonic Distance Sensor
Distance: 41cm



The diagram shows an ESP32 microcontroller board connected to an HC-SR04 ultrasonic sensor. The sensor's VCC pin is connected to the ESP32's 5V pin, and its GND pin is connected to the ESP32's GND pin. The sensor's Trig pin is connected to the ESP32's D4 pin, and its Echo pin is connected to the ESP32's D5 pin. Three LEDs (red, blue, and green) are connected to the ESP32's D0, D1, and D2 pins respectively, each through a 220-ohm resistor. The LEDs are also connected to the sensor's Trig pin.

LED1 + LED2 ON
Data sent to ThingSpeak

Distance: 41.02 cm
ALL LEDs ON
Data sent to ThingSpeak

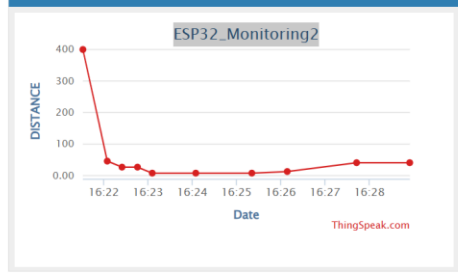
ThingSpeak™ Channels Apps Devices Support Commercial Use How to Buy AS

Channel Stats

Created: 37 minutes ago
Entries: 13

Field 1 Chart

ESP32_Monitoring2



The chart displays distance measurements in centimeters over a period of time. The y-axis is labeled 'DISTANCE' and ranges from 0.00 to 400. The x-axis is labeled 'Date' and shows dates from 16:22 to 16:28. The data points show a sharp drop from approximately 400 cm to near 0 cm, followed by a slight increase and then a steady decline.

Date	Distance (cm)
16:22	400
16:23	50
16:24	20
16:25	10
16:26	5
16:27	10
16:28	50

ThingSpeak.com

LAB Assessment

Student Name		LAB Rubrics	CLO3, P5, PLO5
		Total Marks	10
Registration No		Obtained Marks	
		Teacher Name	Dr. Syed M Hamedoon
Date		Signature	